Why reduce the number of dimensions?

- **A map of ocean temperature has about 16,000 grid cells.**
- **People who forecast El Niño watch one number,** the Niño3.4 index
- **That works because most of the cells warm and cool together.**
- **Dimensionality reduction finds those few numbers from the data itself.**
- **This week's method is principal component analysis, PCA.** Climate scientists call it EOF analysis
- **It is unsupervised.** There is no target. The method only looks at the inputs

Compress
2,469 grid cells per month become 18 numbers. Those 18 keep 90% of the variation.

Discover
The first pattern PCA finds in Pacific SST is El Niño. We never told it where to look.

Prepare
Ten PCs give a better six-month forecast of Niño3.4 than the raw grid cells do.

Week 7 lecture notes on dimensionality reduction, and the tutorial on tropical Pacific SST

Too many inputs is a real problem

- **The task:** predict Niño3.4 six months ahead from this month's SST anomalies
- **Train on 1950 to 2000.** Test on 2001 to 2024
- **Linear regression on all 2,469 cells fits the training years almost perfectly.**
- **On the test years it does worse than guessing zero.**
- **The same regression on 10 principal components beats persistence.**
- **Nothing changed except the number of inputs.**

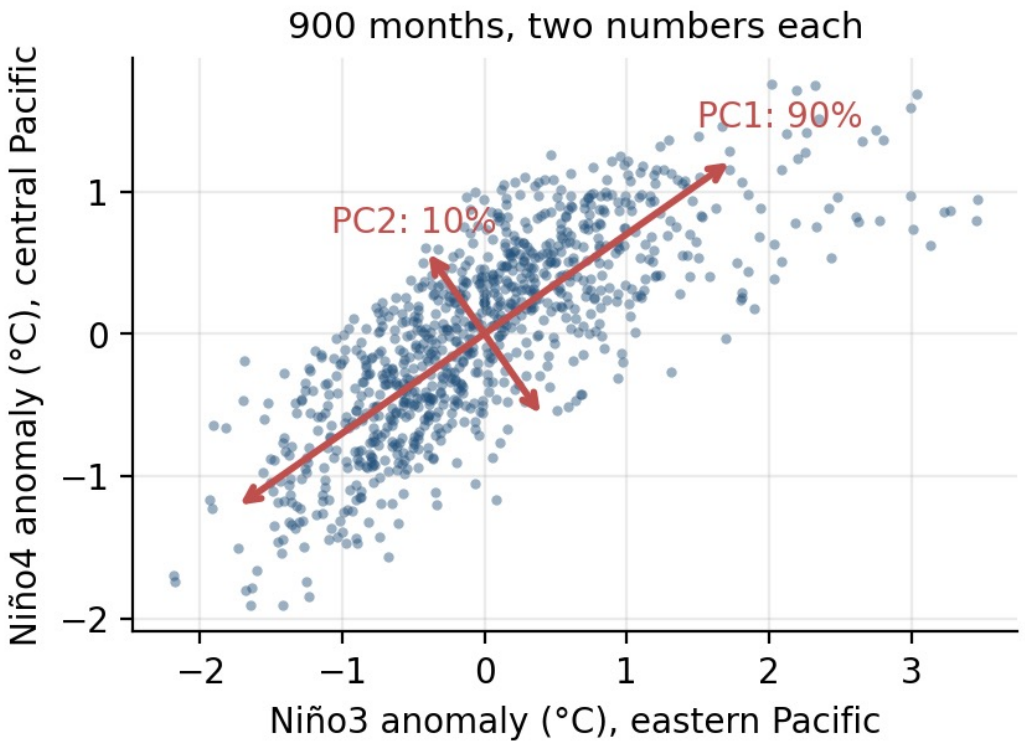
Six months ahead, RMSE (°C)	Train	Test
Climatology, predict zero		0.85
Persistence, this month's value		0.96
Regression on 2,469 grid cells	0.32	1.49
Ridge on 10 principal components		0.74

Why this happens. There are 612 training months and 2,469 inputs. The model has more knobs than data points. It can match the noise in every month exactly, and that noise never repeats.

More data would help, but there is no more. The record is 75 years long. So we need fewer inputs instead.

NOAA ERSST v5 monthly sea surface temperature, tropical Pacific, 1950 to 2024. Test years 2001 to 2024. Persistence means carrying this month's Niño3.4 value forward

What PCA does, in two dimensions



- **Each dot is one month.** The two numbers are the SST anomaly in the eastern Pacific (Niño3) and in the central Pacific (Niño4)
- **The dots form a tilted ellipse.** When the east is warm, the center is usually warm too
- **PCA finds the long axis of the ellipse.** That direction is the first principal component. It holds 90% of the spread
- **The short axis, at right angles, is the second.**
- **Keep only the position along the long axis** and you have one number per month instead of two
- **A full map works the same way,** with 2,469 axes instead of two

NOAA ERSST v5 monthly sea surface temperature, tropical Pacific, 1950 to 2024. Niño3 is 5S to 5N, 150W to 90W. Niño4 is 160E to 150W. Arrows are two standard deviations long

The recipe

- **Put the data in a matrix.** One row per month, one column per grid cell
- **Subtract the mean of each column.** PCA looks at variation around the mean
- **Find the directions of largest spread.** The singular value decomposition does this
- **Project the data onto those directions.** The results are the scores
- **Keep the first few directions and drop the rest.**

Statistics and ML	Climate science	In scikit-learn
principal component (direction)	EOF, the spatial pattern	components_[k]
score (coordinate)	principal component, PC, the time series	transform(X)[:, k]
explained variance	eigenvalue of the covariance	explained_variance_
share of variance	fraction of variance explained	explained_variance_ratio_
loading	loading, weight	entry of components_

```
pca = PCA(n_components=10).fit(X)      # X: months by cells
scores = pca.transform(X)              # months by 10: the PCs
patterns = pca.components_             # 10 by cells: the EOFs
share = pca.explained_variance_ratio_  # fraction of variance each
```

Two names for one method. Statisticians invented PCA in 1901. Meteorologists reinvented it in 1956 and called it EOF analysis. Watch out for the phrase "principal component." In machine learning it means the direction. In climate science it means the time series.

Pearson 1901, Hotelling 1933. Lorenz 1956 for the EOF name. See the lecture notes for the SVD

A little algebra: eigenvectors, orthogonality, SVD

$$C = \frac{X^{\mathrm{T}}X}{n - 1}$$

Start with the covariance matrix. X is the centered data: one row per month, one column per cell. Entry (i, j) of C says how cells i and j vary together.

$$Cv = \lambda v$$

The patterns are its eigenvectors. v is a pattern, the EOF. λ is the variance along it. Sort by λ, largest first, and you have EOF 1, EOF 2, and so on.

$$v_i{}^{\mathrm{T}} v_j = 0 \quad \text{for } i \neq j$$

The patterns are orthogonal. Each one is at right angles to all the others, so the time series are uncorrelated. The method forces this. It is also why the variances add up to the total.

$$X = USV^{\mathrm{T}}$$

The SVD computes everything at once. The columns of V are the patterns. U S holds the time series. The variance of pattern k is $s_k^2 / (n - 1)$. There is no need to build C, which matters when p is large.

In words

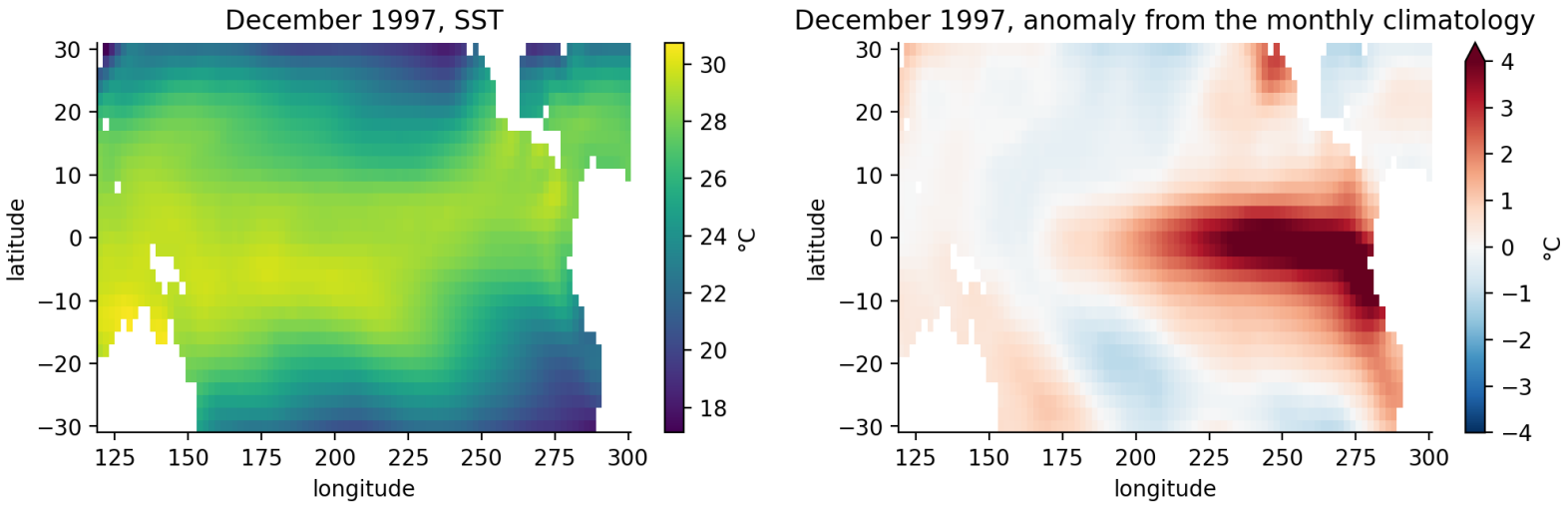
eigenvector: a pattern, the EOF
eigenvalue: the variance that pattern explains
orthogonal: the patterns do not overlap, so the PCs are uncorrelated
SVD: the fast way to compute all of it

Back to the ellipse

On slide 3, PCA found the long and short axes of the ellipse. Those axes are the eigenvectors of C. Their lengths squared are the eigenvalues. Nothing changes with 2,469 axes instead of 2.

The extra slide at the end has the full derivation. Lecture notes, section on PCA

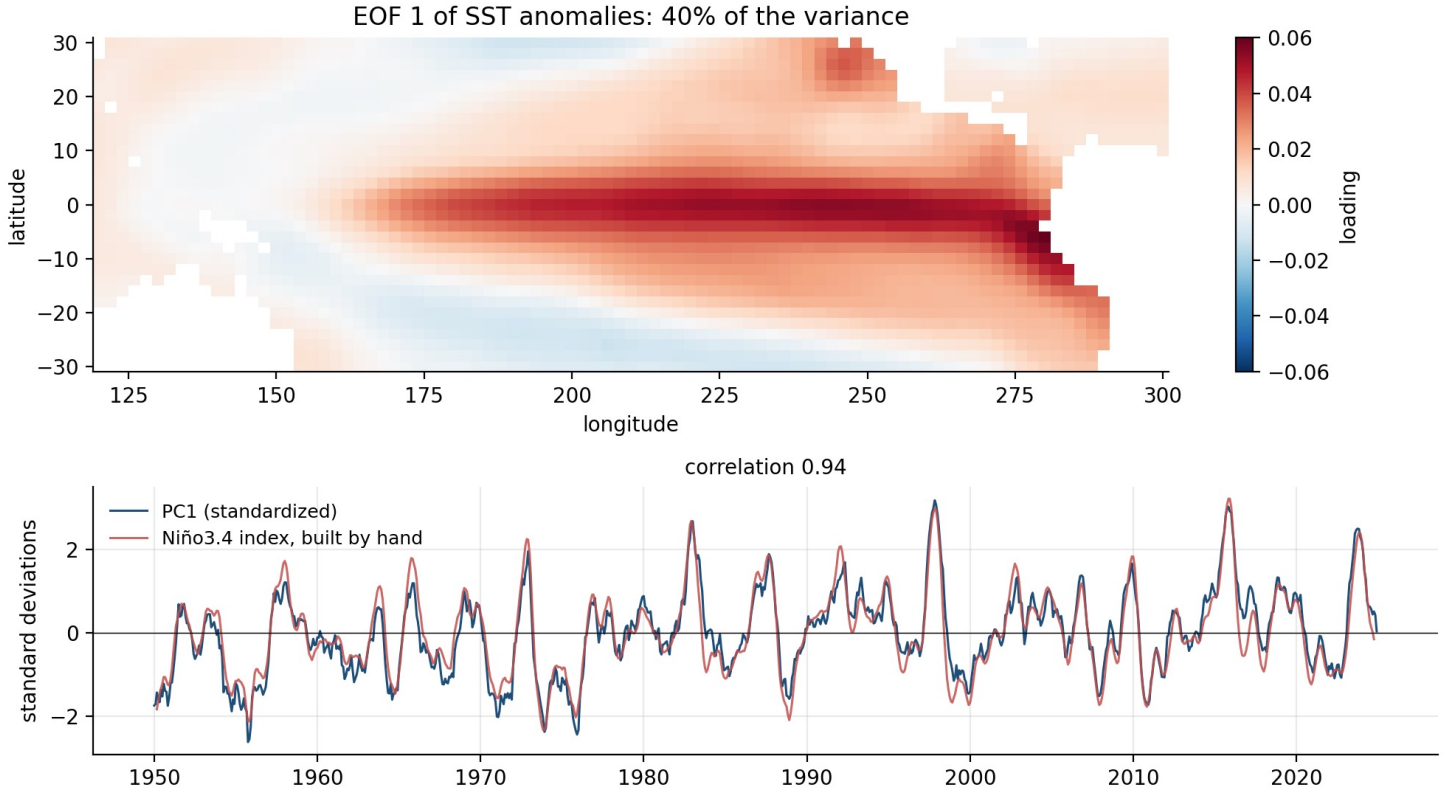
Why sea surface temperature



- **The ocean holds heat for months.** That is why seasonal forecasts start from sea surface temperature
- **El Niño is the biggest year-to-year climate signal on Earth.** It shifts rainfall on every continent
- **The raw map on the left does not show it.** The tropics are warm everywhere
- **The anomaly map on the right does.** This is December 1997 minus an average December. The warm tongue along the equator is El Niño, 3 to 4 degrees above normal
- **Give PCA 900 maps like this and no hints.** Will it find the pattern?

NOAA ERSST v5 monthly sea surface temperature, tropical Pacific, 1950 to 2024. Left: SST. Right: anomaly from the 1950 to 2024 average for December. Land is white

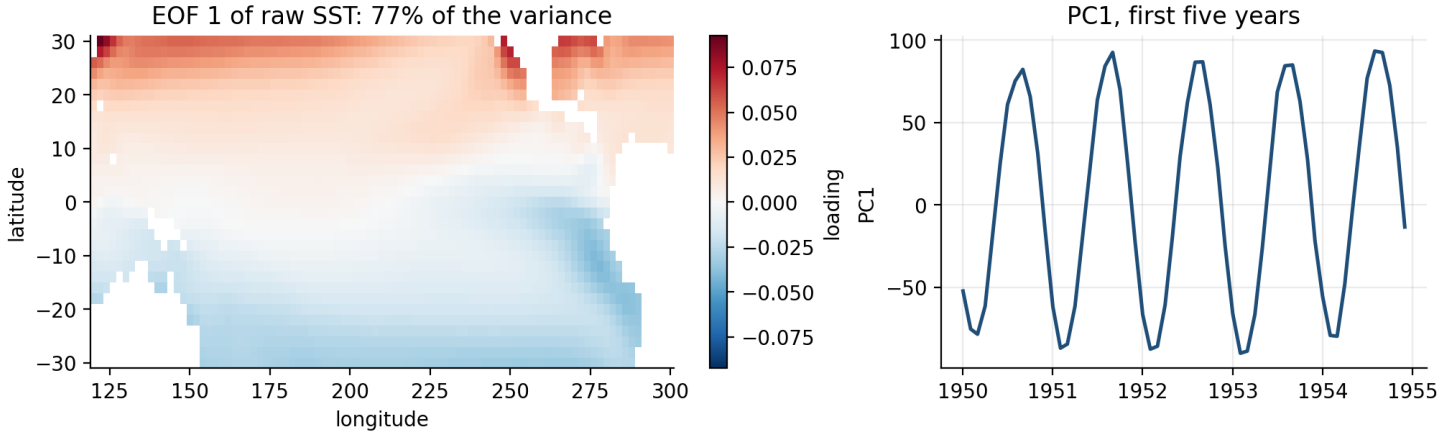
PCA finds El Niño by itself



- **The top map is the first EOF.** It explains 40% of the variation in 2,469 cells
- **It looks like El Niño.** Warm along the equator, strongest in the east
- **The bottom panel is its time series, the first PC.**
- **It matches the Niño3.4 index with a correlation of 0.94.**
- **Scientists built that index by hand from a box on the map.** PCA was given no box
- **It found the pattern because the pattern is where most of the variation is.**

NOAA ERSST v5 monthly sea surface temperature, tropical Pacific, 1950 to 2024. Area weighted, land cells dropped. Niño3.4 index: 5S to 5N, 170W to 120W, 5-month running mean, standardized

Prepare the field before you run PCA



On the raw field, the first pattern is the seasonal cycle. It explains 77% of the variation. That is correct, and not interesting. PCA picks out whatever varies most. Remove what you already know about before you run it.

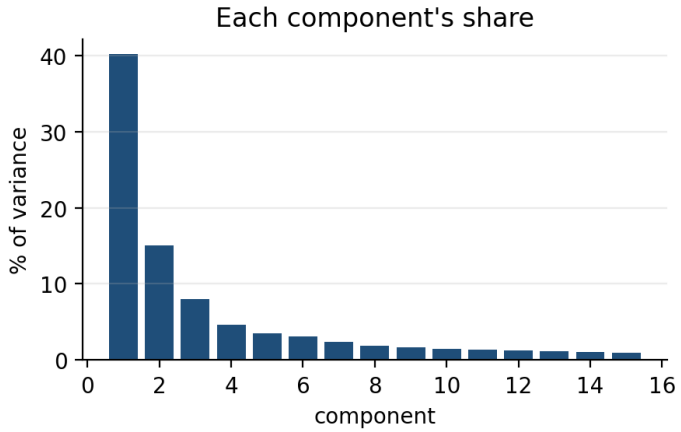
- **Use anomalies.** Subtract the average for each calendar month. Remove the trend too, unless the trend is what you want to study
- **Weight by area.** Grid cells near the poles are smaller than cells at the equator. Multiply each cell by the square root of the cosine of its latitude
- **Drop the land cells.** Keep a list of which cells are ocean and run PCA on those. Do not fill land with zeros
- **Do not standardize the cells.** Scaling every cell to the same variance erases the signal. Center the data, but do not scale it

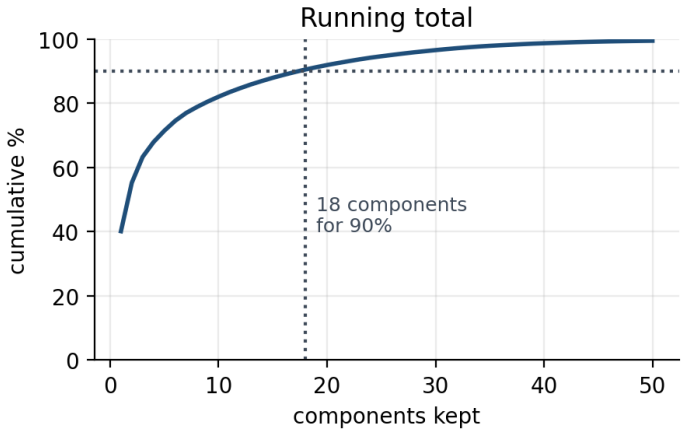
The reshape is most of the work. A (time, lat, lon) array becomes a (time, cells) matrix. Land columns come out with a mask. Each EOF goes back onto the map with the same mask. The tutorial's to_matrix and to_map functions do this.

Standardizing is right when the columns are different variables in different units, like temperature next to pressure. It is wrong when every column is the same variable.

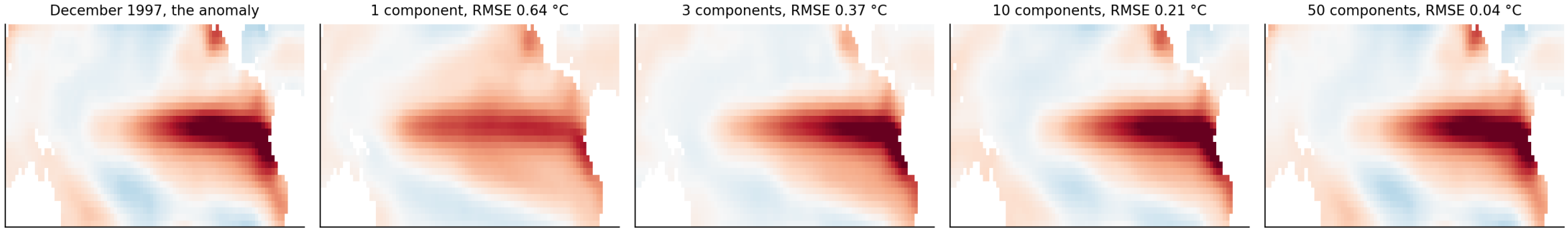
NOAA ERSST v5 monthly sea surface temperature, tropical Pacific, 1950 to 2024, before removing the seasonal cycle. PC1 shown for 1950 to 1954

How many components to keep





- **The spectrum shows how much each component explains.** It drops fast, then flattens
- **The first component has 40%.** The first ten have 82%. Reaching 90% takes 18
- **PCA(n_components=0.9)** keeps enough for 90%. The 90 is a habit, not a rule
- **Rebuild a map from a few components and look at what is missing.** December 1997 from 10 components is close to the original. The error is 0.21 °C



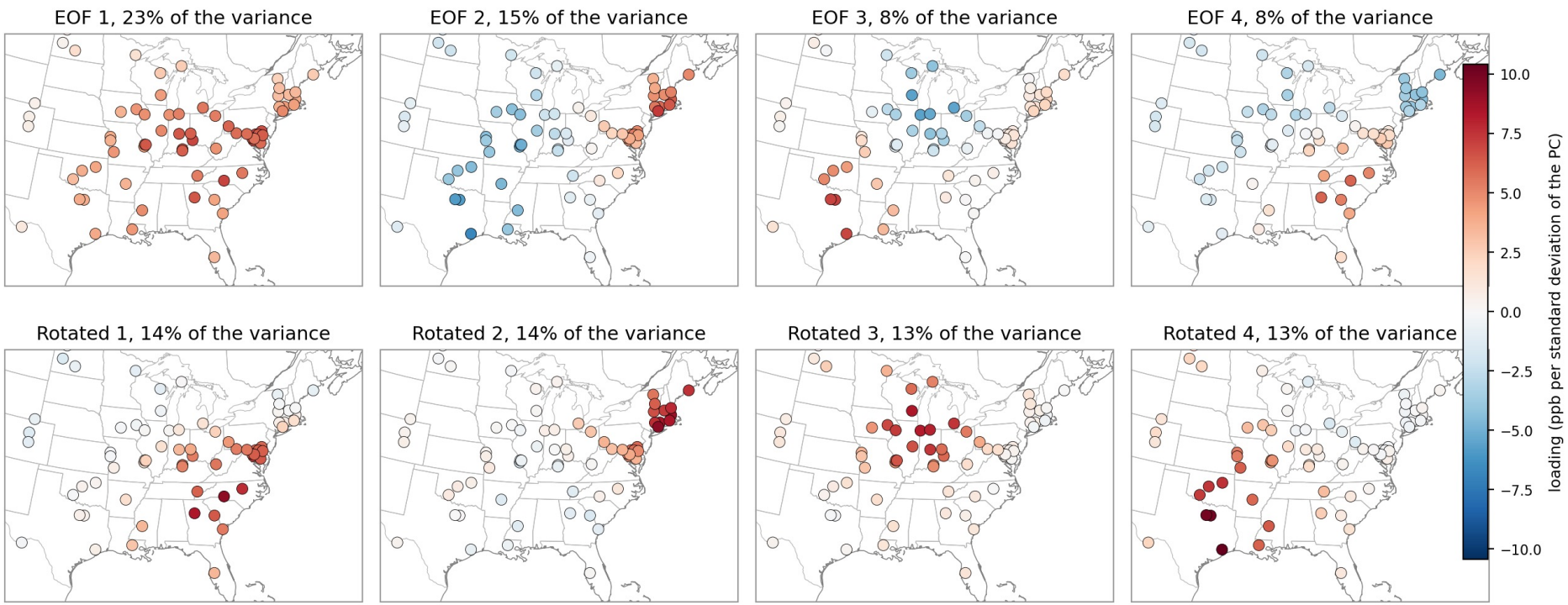
NOAA ERSST v5 monthly sea surface temperature, tropical Pacific, 1950 to 2024. Bottom row: the December 1997 anomaly rebuilt from the first k components, same color scale. RMSE over ocean cells

Reading the patterns carefully

- **The sign is arbitrary.** A pattern and its negative are the same pattern. Pick a sign convention and say what it is
- **The patterns are forced to be at right angles.** The real climate system is not. So EOF 2 is not automatically a second physical mode
- **If two components explain almost the same amount, PCA mixes them.** Treat them as a pair
- **Biggest is not the same as most important.** A small component can matter most for your question
- **An EOF cannot move.** It only grows and shrinks. A traveling wave needs two EOFs to describe it
- **The pattern depends on the region you chose.** Change the box and the pattern changes

North et al. 1982 for the rule on close eigenvalues. Hannachi, Jolliffe and Stephenson 2007 for a review of EOFs in atmospheric science

Rotated EOFs: patterns that are easier to read



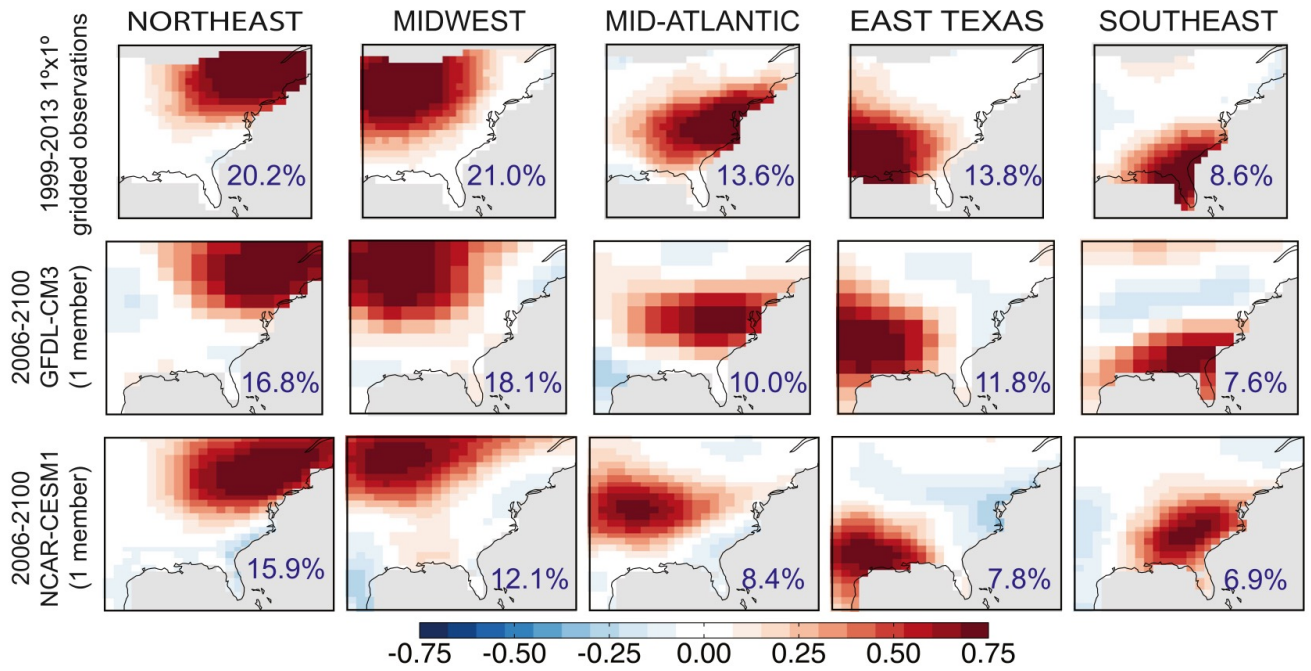
- **Top row: plain EOFs of summer ozone in 69 eastern counties.** EOF 1 is everything high at once. EOF 2 is the Northeast against the rest
- **Bottom row: the same four EOFs after a varimax rotation.** Each one is now a single region: the Southeast, New England, the Midwest, Texas and the Gulf
- **The total variance kept is the same, 54%.** Rotation only shares it out differently among the four

What rotation does. Take the first k EOFs and turn them within the space they span. Pick the angle that makes each pattern as concentrated as possible: a few large loadings, the rest near zero. Varimax is the usual rule for that.

What you give up. The ordering by variance. One kind of orthogonality: either the patterns or the time series stay uncorrelated, not both. And the answer depends on k. Rotate four and you get four regions. Rotate six and you get six.

Daily ozone from the Week 6 data set, May to September 2019 to 2023, counties with at least 80% of days reported, anomalies from each county's monthly mean. Varimax on the first four EOFs

Application: pollution episodes in a warming climate



Why rotate here

The regions come from the data, not from state lines. The same five show up in the observations and in both models. That is also how the models were checked: a free-running model cannot match the weather on a given day, but it can match the patterns.

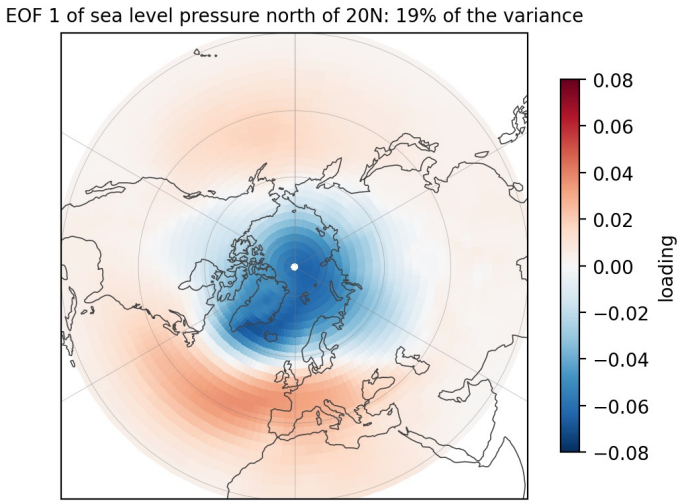
Why an index and not a threshold

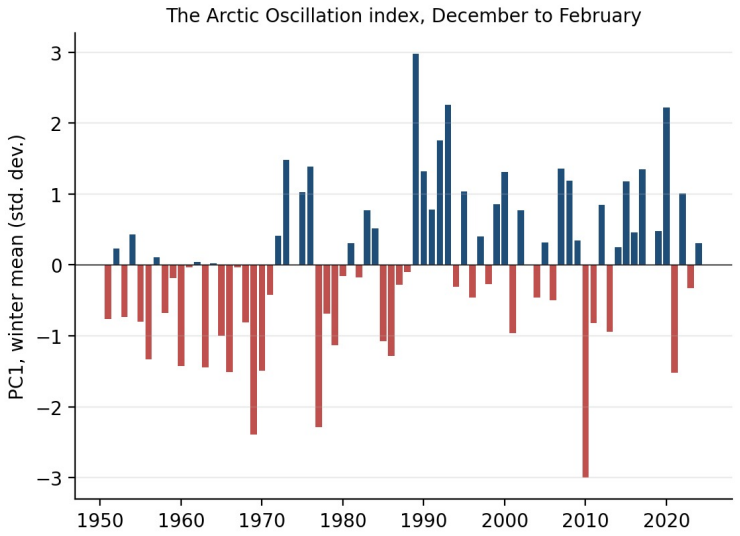
Both models run too high in PM2.5. A concentration threshold would need a separate correction for each model and region. The top quarter of a PC comes from each model's own variability, so nothing needs correcting.

- **The question:** do summer pollution episodes over the eastern US get more frequent or longer as the climate warms, with emissions held fixed?
- **The data:** daily PM2.5, ozone, and temperature, June to August. Gridded observations for 1999 to 2013, and two climate models run to 2100 under a high warming scenario with air pollutant emissions frozen at 2005 levels. Three runs of GFDL-CM3, twelve of NCAR-CESM1
- **The method:** standardize each grid cell, keep the first five EOFs, rotate them with varimax. The five rotated patterns are the five regions on the left. Each PC is a daily index for its region
- **An episode:** three or more days in a row with the index in its top quarter. Count them per decade. The spread across runs is the noise a change has to beat
- **The result:** Northeast summers warm by more than 4 °C by 2081 to 2100. Multi-day Northeast PM2.5 episodes double per decade in GFDL-CM3 and rise by about half in CESM1. Elsewhere the two models disagree by more than the noise

Figure 1 of Fiore, Milly, Hancock, et al. 2022, J. Geophys. Res. Atmospheres, 127, e2021JD035985, open access CC BY-NC. Varimax-rotated EOFs of standardized summer daily PM2.5. Blue numbers: share of variance

Other climate patterns found the same way





- **Many named climate patterns are the first EOF of some field over some region.**
- **Arctic Oscillation:** first EOF of sea level pressure north of 20N, shown here. Low over the pole, high at mid-latitudes. It shapes winter in Europe and the eastern United States
- **North Atlantic Oscillation:** the same idea, Atlantic sector only
- **Southern Annular Mode:** the Antarctic version
- **Pacific Decadal Oscillation:** first EOF of North Pacific SST. This week's assignment
- **Madden-Julian Oscillation:** a pattern that moves, so it uses the first two EOFs together

NCEP-NCAR Reanalysis 1 monthly sea level pressure, 20N to 87.5N, 1950 to 2024, anomalies, area weighted. Right: winter mean of PC1, December to February

PCA as a first step before a model

Seasonal forecasts
A few PCs of SST are the inputs to a regression. Our ten PCs beat persistence at six months.

Climate model emulators
Compress the model's output map to a few dozen PCs. Fit the emulator to those. Rebuild the map. Week 13.

Weather forecasting
Ensemble forecasts and data assimilation work in a smaller space built from EOFs of the forecast errors.

Past climates
Project tree rings, corals, and ice cores onto modern EOFs to rebuild temperature maps from before the instruments.

Air quality
PCA on daily aerosol chemistry. Each component is a source: traffic, dust, sea salt, wood smoke.

Satellite data
Hundreds of channels compressed to a few before classification.

- **The pattern is the same each time.** Unsupervised first, then supervised
- **The Week 6 rules apply.** Fit PCA on the training years only. Put it in the Pipeline so cross-validation refits it in each fold
- **An autoencoder, in Week 12,** learns a nonlinear version of the same compression

Forecast numbers from the Week 7 tutorial, section 9. Pipeline leakage: Week 6 slide on leakage through preprocessing

When PCA is not the right tool

- **The data lie on a curve, not a line.** Kernel PCA, manifold methods, or autoencoders
- **The pattern moves.** Extended or complex EOFs, or a model with time in it, like the RNN in Week 11
- **You want patterns that are local and easy to read.** Rotated EOFs, slide 11
- **The amounts must be positive, like pollution sources.** Non-negative matrix factorization
- **You want a picture of the data, not a model.** t-SNE and UMAP, in Week 9
- **You have a target and want directions that predict it.** Partial least squares

Check the preparation first

Did you remove the seasonal cycle? Weight by area? Check that the leading eigenvalues are separated? Most strange EOFs come from the preparation, not from the method.

This week

Tutorial: PCA on tropical Pacific SST, and the forecast from PCs. Assignment: the Pacific Decadal Oscillation. Next week: clustering, the other half of unsupervised learning.

Lecture notes, sections on when PCA does not work well and where next. Hannachi et al. 2007 for rotated and extended EOFs

Extra

The linear algebra behind PCA

$$C = \frac{1}{n-1} X^{\mathrm{T}} X$$

Covariance of the centered data. X is n by p: n months, p cells.

$$T = XV = US$$

Scores. Column k of T is PC k, the time series of EOF k.

$$CV = V\Lambda, \qquad V^{\mathrm{T}}V = I$$

Eigen-decomposition. The columns of V are the EOFs. Λ is diagonal, with $\lambda_1 \geq \lambda_2 \geq \ldots$ on the diagonal. $V^{\mathrm{T}}V = I$ says the EOFs are orthonormal.

$$\hat{X}_k = U_k S_k {V_k}^{\mathrm{T}} = \sum_{j=1}^{k} t_j {v_j}^{\mathrm{T}}$$

Rebuild from the first k. Each term is one time series times one map.

$$v_1 = \underset{\|v\|=1}{\operatorname{argmax}}\, v^{\mathrm{T}} C v$$

The first EOF has the largest variance, which is λ_1. Each next one does the same, at right angles to the earlier ones.

$$\left\| X - \hat{X}_k \right\| \leq \|X - B\|$$

No other k patterns do better. For any other matrix B of rank k. Eckart and Young, 1936. This is why PCA is the default for compression.

$$X = USV^{\mathrm{T}}, \qquad \lambda_k = \frac{s_k^2}{n-1}$$

The SVD gives the same V without ever forming C. That matters when p is large.

$$\frac{\lambda_k}{\sum_j \lambda_j}, \qquad \sum_j \lambda_j = \operatorname{tr} C$$

Share of variance. The λ add up to the total variance because the EOFs are orthogonal.

```
Xc = X - X.mean(axis=0)                            # center
U, s, Vt = np.linalg.svd(Xc, full_matrices=False)  # SVD
eofs, pcs = Vt, U * s                              # patterns, time series
var = s**2 / (len(X) - 1); share = var / var.sum()
```

scikit-learn's PCA is this SVD, with the centering done for you. The results match up to the sign of each component.

With n < p, C has rank at most n − 1. So 900 months give at most 899 EOFs, not 2,469. Area weights multiply the columns of X before the SVD.

Lecture notes, section on PCA. Jolliffe, Principal Component Analysis, for the rest

PCA in scikit-learn, in one place

```
from sklearn.decomposition import PCA

pca = PCA(n_components=10)          # or 0.9 for "enough for 90%"
pca.fit(X_train)                     # centers, then SVD; train rows
T_train = pca.transform(X_train)     # scores, n_train by 10
T_test  = pca.transform(X_test)      # same patterns applied to new rows

pca.components_                      # 10 by p: the EOFs
pca.explained_variance_ratio_        # share of variance, sums to <= 1
pca.n_components_                    # how many were kept
pca.mean_                            # the column means it subtracted
X_hat = pca.inverse_transform(T_test)   # rebuild from 10

# inside a supervised pipeline, refit in every fold
model = make_pipeline(PCA(n_components=10), Ridge(alpha=1.0))
cross_val_score(model, X, y, cv=TimeSeriesSplit(5))
```

- **Fit on the training rows only,** then transform both sets. The components are learned, like a scaler's mean
- **n_components** takes a whole number, or a fraction between 0 and 1 for a variance target
- **Large matrices** switch to a randomized solver automatically. IncrementalPCA works in batches when X does not fit in memory
- **A trailing underscore** marks something that fit learned
- **The sign of each component** can differ between versions and solvers. Fix it yourself before comparing to a published index

scikit-learn 1.7, sklearn.decomposition.PCA. The pipeline line is the Week 6 leakage rule applied to PCA